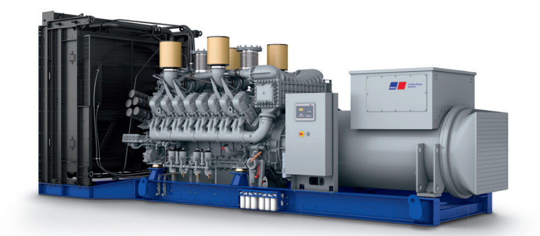
Diesel Generator Set

mtu 16V4000 DS2750

380V – 11 kV/50 Hz/prime power for stationary emergency/ NEA (ORDE) + Tier 2 optimized/16V4000G34F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
— Low fuel consumption
— Optimized system integration ability
— High reliability
— High availability of power
— Long maintenance intervals

Support
— Global product support offered

Standards
— Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
— Generator set complies to ISO 8528
— Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
— NFPA 110

Power rating
— System ratings: 2470 kVA - 2600 kVA
— Accepts rated load in one step per NFPA 110*
— Generator set complies to G3 according to ISO 8528-5
— Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
— Engine-generator set tested to ISO 8528-5 for transient response
— 85% load factor
— Verified product design, quality and performance integrity
— All engine systems are prototype and factory tested

Complete range of accessories available
— Control panel
— Power panel
— Circuit breaker/power distribution
— Fuel system
— Fuel connections with shut-off valve mounted to base frame
— Starting/charging system
— Exhaust system
— Mechanical and electrical driven radiators
— Medium and oversized voltage alternators

Emissions
— Tier 2 optimized engine
— NEA (ORDE) optimized engine

Certifications
— CE certification option
— Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1]

Engine

Manufacturer	*mtu*
Model	16V4000G34F
Type	4-cycle
Arrangement	16V
Displacement: l	76.3
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ADEC (ECU 9)
Max power: kWm	2387
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption [2]	l/hr	g/kwh
At 100% of power rating:	561	195
At 75% of power rating:	430	199
At 50% of power rating:	297	206

Liquid capacity (lubrication)

Total oil system capacity: l	300
Engine jacket water capacity: l	175
Intercooler coolant capacity: l	50

Combustion air requirements

Combustion air volume: m^3/s	2.7
Max. air intake restriction: mbar	30

Cooling/radiator system

Coolant flow (HT-circuit) at 0,3 bar: m^3/hr	63
Coolant flow (HT-circuit) at 0,7 bar: m^3/hr	53
Coolant flow (NT-circuit) at 0,3 bar: m^3/hr	33
Coolant flow (NT-circuit) at 0,7 bar: m^3/hr	25
Heat rejection to coolant: kW	785
Heat radiated to charge air cooling: kW	505
Heat radiated to ambient: kW	90

Exhaust system

Exhaust gas temp. (after engine): °C	450
Exhaust gas temp., max (after engine): °C	550
Exhaust gas temp. (before turbocharger): °C	680
Exhaust gas volume: m^3/s	6.8
Maximum allowable back pressure: mbar	50

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) + Tier 2 optimized					
		without radiator			with radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 UL16 (Low voltage Leroy Somer standard)	380 V	2080	2600	3950	2008	2510	3814
	400 V	2080	2600	3753	2008	2510	3623
	415 V	2080	2600	3617	2008	2510	3492
Leroy Somer LSA53.2 M9 (Low voltage Leroy Somer oversized)	380 V	2080	2600	3950	2016	2520	3829
	400 V	2080	2600	3753	2016	2520	3637
	415 V	2080	2600	3617	2016	2520	3506
Marathon 1020FDL7108 (Low voltage Marathon)	380 V	2080	2600	3950	1976	2470	3753
	400 V	2080	2600	3753	1976	2470	3565
	415 V	2080	2600	3617	1976	2470	3436
Leroy Somer LSA 53.2 XL11 (Medium volt. Leroy Somer)	11 kV	2080	2600	136	2008	2510	132
Marathon 1030FDH7100 (Medium volt. Marathon)	11 kV	2032	2540	133	2008	2510	132

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.